

Eunbi Yoon

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Education

University of Massachusetts Amherst, Sep 2024 - Dec 2026 | M.S. in Computer Science, AI Concentration | GPA: 3.93

- Systems for Data Science, Machine Learning, Neural Networks, Advanced Text as Data, Artificial Intelligence

Kyung Hee University, Mar 2016 – Feb 2020 | B.E. in Electronics Engineering, Software Engineering Track

- Object Oriented Programming, Programming Structures, Computer Architecture, Digital Signal Processing

Certifications

- AWS Certified Machine Learning Engineer – Associate (MLA-C01) — Amazon Web Services, Sep 2026
- Deep Learning with TensorFlow – Advanced (ML0120EN) — IBM, Sep 2026
- AI Agent Fundamentals & Generative AI Fundamentals — Databricks, Jun 2026

Skills

Languages/Cloud/Tools: Python, Java, C#, SQL, R, AWS, Docker, Git, Linux, Spark/PySpark, W&B, TensorBoard

Backend & Data: REST APIs, FastAPI, Flask, Django, SQL, NoSQL, Data Pipelines, API Integration

AI/ML: PyTorch, TensorFlow, LLM APIs, Embeddings, RAG, Vector Databases, Model Evaluation, Recommendation Systems

LLM & GenAI: Hugging Face Transformers, LangChain, LlamaIndex, LoRA, Fine-Tuning

Professional Experiences (5+ years)

PTKOREA (Samsung Subsidiary), Gangnam, Seoul, South Korea – Samsung.com Data Engineering Team

Backend Engineer 2 (Full-time), Feb 2025 – Oct 2025

- Developed traffic data pipelines for Samsung.com, ingesting billion-scale traffic data from third-party Adobe Analytics APIs, preprocessing with Spark and SQL, and orchestrating ETL workflows using Kubernetes, Docker, and Airflow on AWS.
- Led backend infrastructure on the internal use finance platform by designing Fast API, Django, PostgreSQL, and REST APIs, deploying backend server in Ubuntu environments with nginx and apache, and collaborating in agile/scrum environment.

LG Electronics North America, Clarksville, TN, United States – R&D Smart IT Team

Engineer 2 (Full-time), Jan 2020 – Jun 2024

- Developed employee time-off and security management application using REST APIs, Flask, Fast API, Django; built scalable backend services and integrated with enterprise systems to ensure reliable synchronization and workflow automation.
- Developed product defect detection computer vision models with TensorFlow and OpenCV, applying preprocessing and class imbalance handling, and validating with precision/recall metrics to improve automated inspection reliability.

Papers / Projects

TRACE: Separating Action and Reasoning Teachers for Strategic SLM Training (with Amazon Science, submitted to NeurIPS 2026 SLM Agent Workshop)

- Developed A+ β , a solver-guided Direct Preference Optimization framework that combines analytical game-theoretic solvers, counterfactual horizon rollouts, and LoRA adapter merging to improve strategic reasoning in 3B language models.
- Improved held-out cumulative reward from -1.45 to +1.91 (93% of the gap to Haiku 4.5), increased tit-for-tat reward from 1.50 to 14.06, and matched frontier-level performance across a 12-environment strategic game benchmark.

Multimodal Small-Object Detection with Faster R-CNN & CLIP

- Built end-to-end multimodal object detection pipeline integrating Faster R-CNN with a ResNet-50-FPN backbone and frozen CLIP for language-guided small-object detection and semantic reranking in highly cluttered scenes.
- Conducted systematic ML model evaluation using Mean IoU and Top-1@0.5 evaluation metrics, comparing model performance across query formulations, training configurations, and attribute-aware learning strategies.

Document-Grounded RAG QA Assistant

- Built an end-to-end document-grounded RAG assistant using LangGraph, LlamaIndex, Chroma, and LLM API(Ollama, OpenAI), implementing document chunking, embedding-based retrieval, and citation-grounded response generation.
- Evaluated RAG performance across retrieval quality, hallucination, and citation grounding, using failure analysis to improve similarity filtering, source-controlled citations, and abstention for insufficient evidence.